

Configuring a VPN Client for Secure File Transfers (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 09

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Time on Task:

18 hours, 45 minutes

Progress:

100%

Report Generated: Wednesday, July 27, 2022 at 10:55 AM

Section 1: Hands-On Demonstration

Part 1: Configure a Windows VPN Client

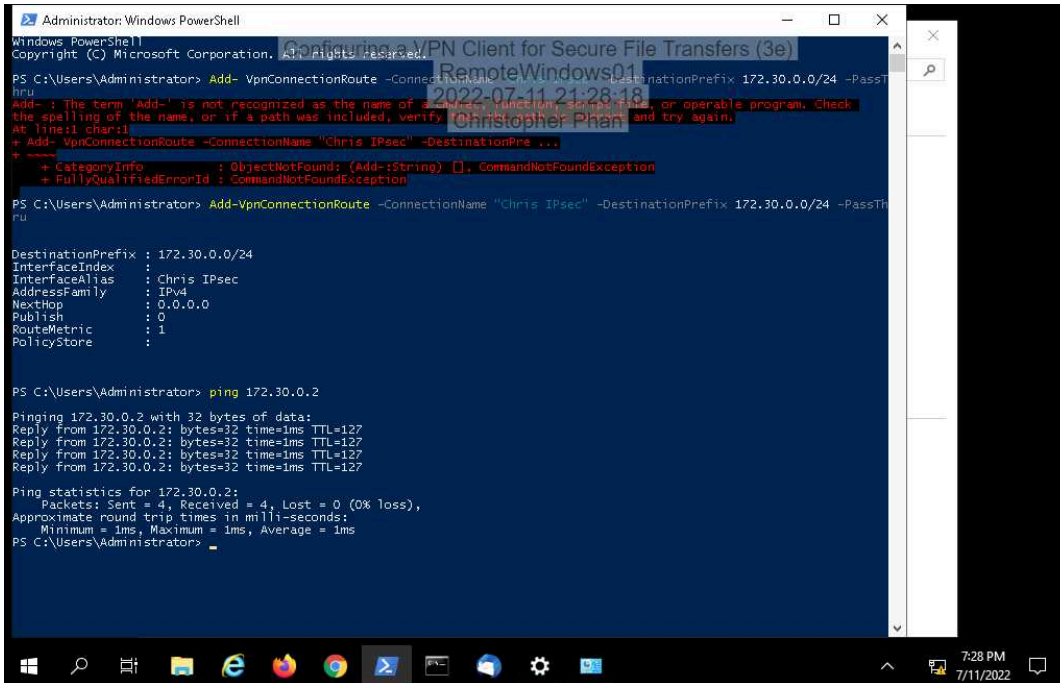
30. Make a screen capture showing the VPN connection properties.



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54. Make a screen capture showing the successful ping response.



```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName "Chris IPsec" -DestinationPrefix 172.30.0.0/24 -PassThru
Add- : The term 'Add-' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ Add-VpnConnectionRoute -ConnectionName "Chris IPsec" -DestinationPne ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Add-:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

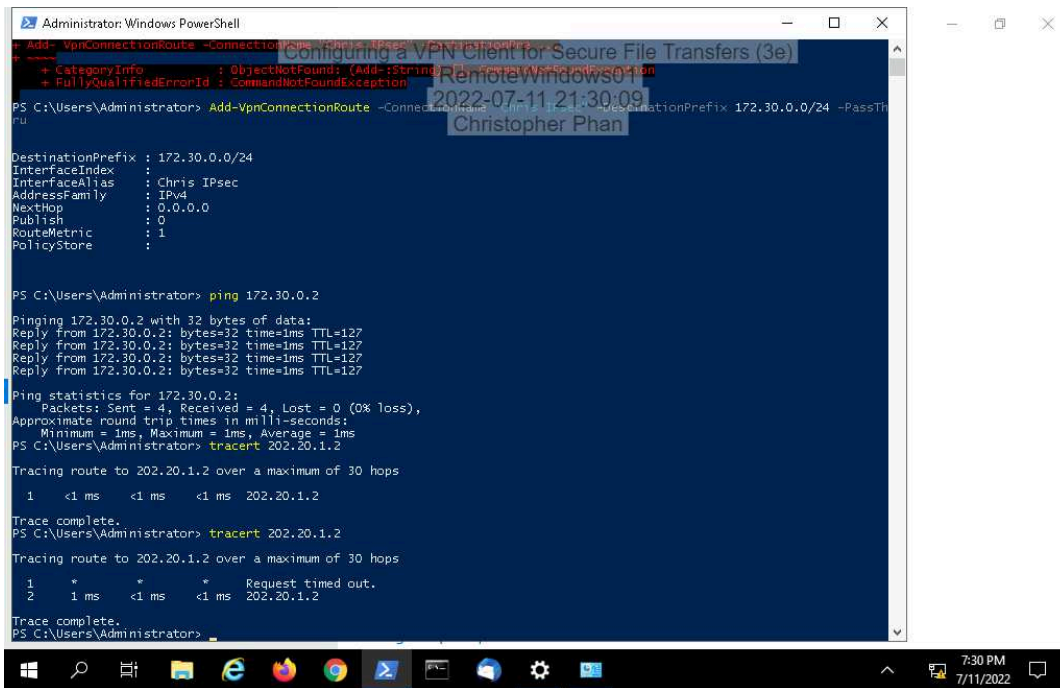
PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName "Chris IPsec" -DestinationPrefix 172.30.0.0/24 -PassThru
DestinationPrefix : 172.30.0.0/24
InterfaceIndex    : 
InterfaceAlias    : Chris IPsec
AddressFamily     : IPv4
NextHop           : 0.0.0.0
Publish           : 0
RouteMetric       : 1
PolicyStore       : 

PS C:\Users\Administrator> ping 172.30.0.2

Pinging 172.30.0.2 with 32 bytes of data:
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127

Ping statistics for 172.30.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\Administrator>
```

72. Make a screen capture showing your new tracert results.



```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName "Chris IPsec" -DestinationPrefix 172.30.0.0/24 -PassThru
Add- : The term 'Add-' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ Add-VpnConnectionRoute -ConnectionName "Chris IPsec" -DestinationPne ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Add-:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> Add-VpnConnectionRoute -ConnectionName "Chris IPsec" -DestinationPrefix 172.30.0.0/24 -PassThru
DestinationPrefix : 172.30.0.0/24
InterfaceIndex    : 
InterfaceAlias    : Chris IPsec
AddressFamily     : IPv4
NextHop           : 0.0.0.0
Publish           : 0
RouteMetric       : 1
PolicyStore       : 

PS C:\Users\Administrator> ping 172.30.0.2

Pinging 172.30.0.2 with 32 bytes of data:
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127
Reply from 172.30.0.2: bytes=32 time=1ms TTL=127

Ping statistics for 172.30.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\Administrator> tracert 202.20.1.2

Tracing route to 202.20.1.2 over a maximum of 30 hops
  0  <1 ms  <1 ms  <1 ms  202.20.1.2
Trace complete.
PS C:\Users\Administrator> tracert 202.20.1.2

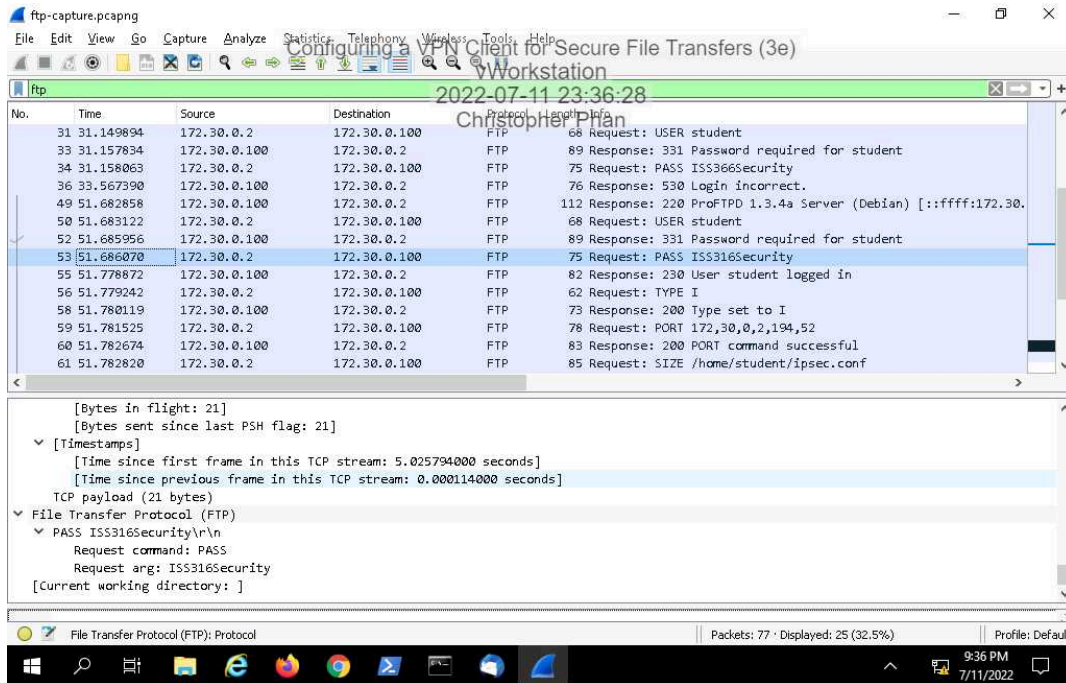
Tracing route to 202.20.1.2 over a maximum of 30 hops
  0  *        *        *        Request timed out.
  1  1 ms     <1 ms    <1 ms    202.20.1.2
Trace complete.
PS C:\Users\Administrator>
```

Part 2: Compare Secure and Non-Secure File Transfers in Wireshark

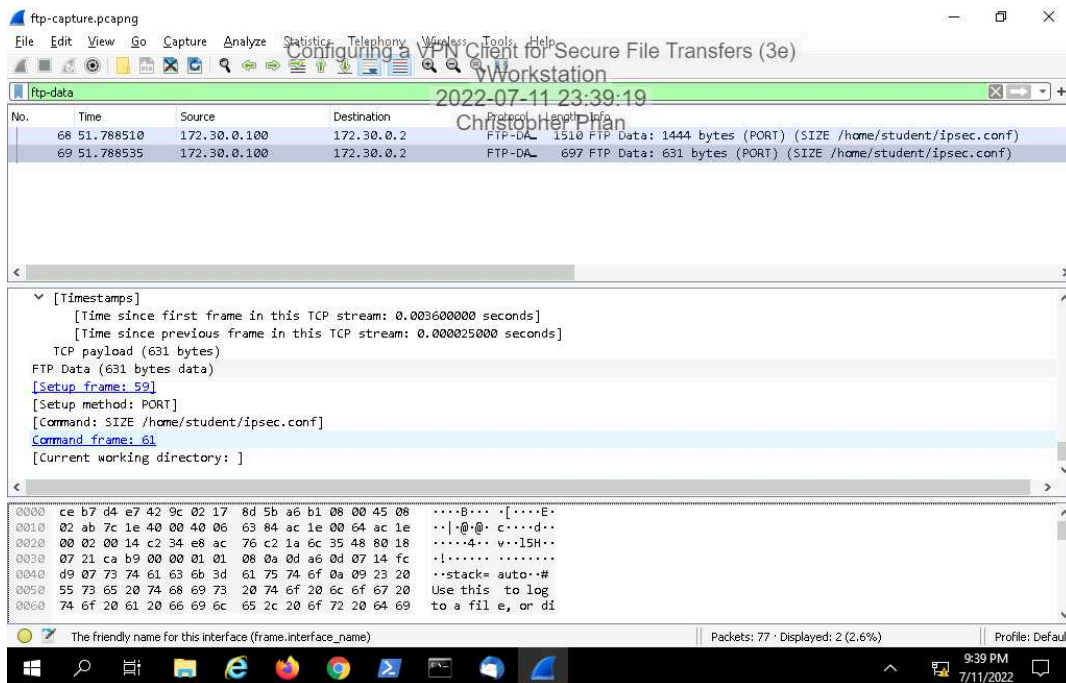
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12. Make a screen capture showing the packet that carries the correct password.



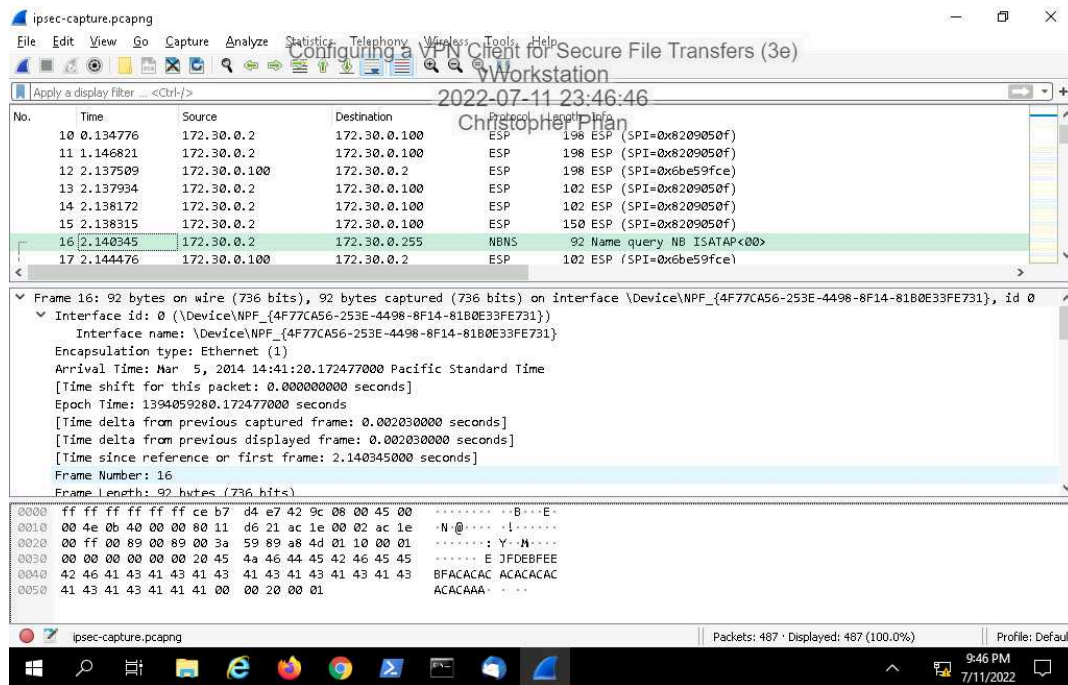
28. Make a screen capture showing the Wireshark window and the packet bytes pane for Packet 69.



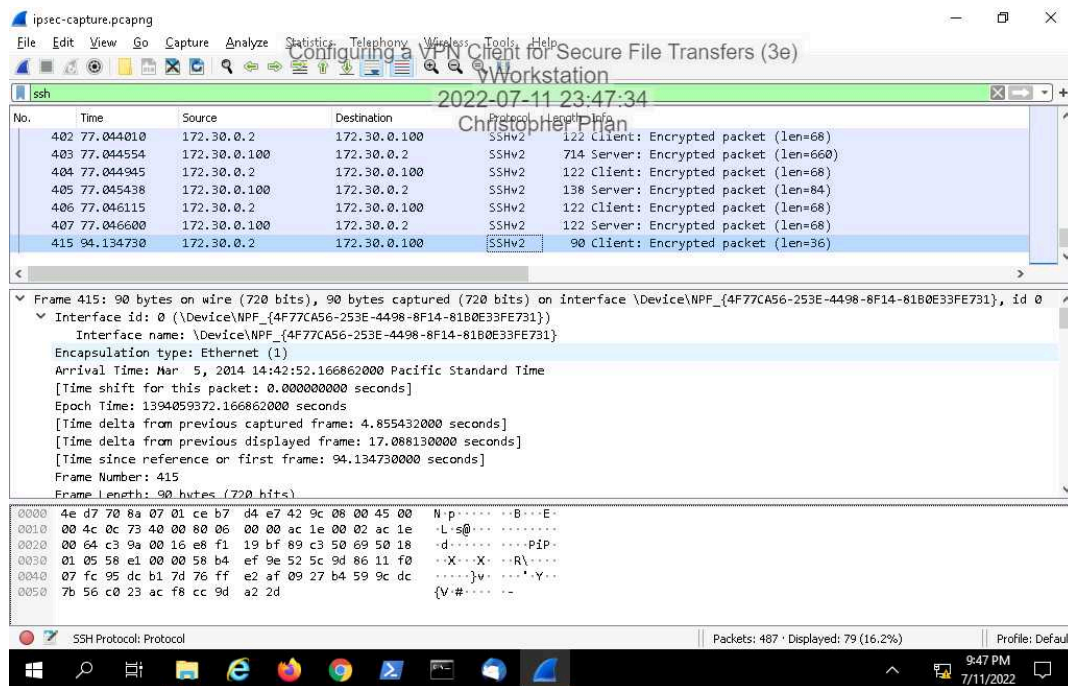
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44. Make a screen capture showing the packet details pane for packet 16.



49. Make a screen capture showing the last SSHv2 packet in the SSH file transfer.



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51. Make a screen capture showing the last packets in the ESP exchange.

The screenshot shows the Wireshark network protocol analyzer interface. The title bar indicates the file is 'ipsec-capture.pcapng'. The main display area shows a list of captured packets, with the last packet (No. 438) selected. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
430	114.287386	172.30.0.2	172.30.0.100	ESP	102	ESP (SPI=0x8209050f)
431	116.964623	172.30.0.2	172.30.0.100	ESP	118	ESP (SPI=0x8209050f)
432	116.970122	172.30.0.100	172.30.0.2	ESP	102	ESP (SPI=0x6be59fce)
435	117.156132	172.30.0.2	172.30.0.100	ESP	118	ESP (SPI=0x8209050f)
436	117.160795	172.30.0.100	172.30.0.2	ESP	102	ESP (SPI=0x6be59fce)
437	117.161026	172.30.0.2	172.30.0.100	ESP	118	ESP (SPI=0x8209050f)
438	117.164619	172.30.0.100	172.30.0.2	ESP	102	ESP (SPI=0x6be59fce)

The packet details pane for the selected packet (No. 438) shows the following information:

- Frame 438: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface \Device\NPF_{4F77CA56-253E-4498-8F14-81B0E33FE731}, id 0
- Interface id: 0 (\Device\NPF_{4F77CA56-253E-4498-8F14-81B0E33FE731})
- Interface name: \Device\NPF_{4F77CA56-253E-4498-8F14-81B0E33FE731}
- Encapsulation type: Ethernet (1)
- Arrival Time: Mar 5, 2014 14:43:15.196751000 Pacific Standard Time
- [Time shift for this packet: 0.000000000 seconds]
- Epoch Time: 1394059395.196751000 seconds
- [Time delta from previous captured frame: 0.003593000 seconds]
- [Time delta from previous displayed frame: 0.003593000 seconds]
- [Time since reference or first frame: 117.164619000 seconds]
- Frame Number: 438

The packet bytes pane shows the raw data in hexadecimal and ASCII format:

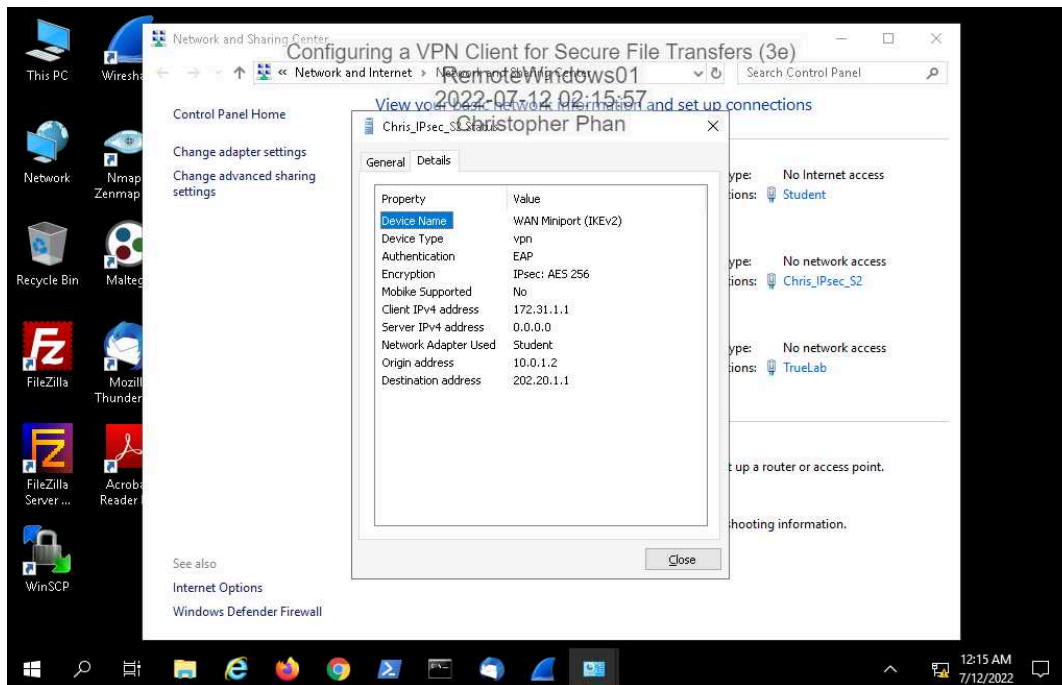
```
0000 ce b7 d4 e7 42 9c 4e d7 70 8a 07 01 08 00 45 00  ....B.N.p....E-
0010 00 58 00 00 40 00 40 32 e1 d1 ac 1e 00 64 ac 1e  X..@.2.....
0020 00 02 6b e5 9f ce 00 00 00 29 8f 0e 04 f2 9e b7  -k.....).....
0030 bd 7c 5a 9c 05 54 d7 c5 db 56 4b 49 96 f9 19 43  -|Z..T...VKI...C
0040 fa 8f 10 d6 d1 13 c3 38 16 3b 83 15 34 45 ec 63  .....8...4E.c
0050 79 5f f6 d3 67 18 23 c2 59 e6 69 3a 65 d9 bd f8  y..g.#.Y.i:..
0060 c2 96 a6 d4 4f 9a  ....O.
```

The status bar at the bottom indicates 'Encapsulating Security Payload: Protocol' and 'Packets: 487 · Displayed: 204 (41.9%)'. The system clock shows 9:47 PM on 7/11/2022.

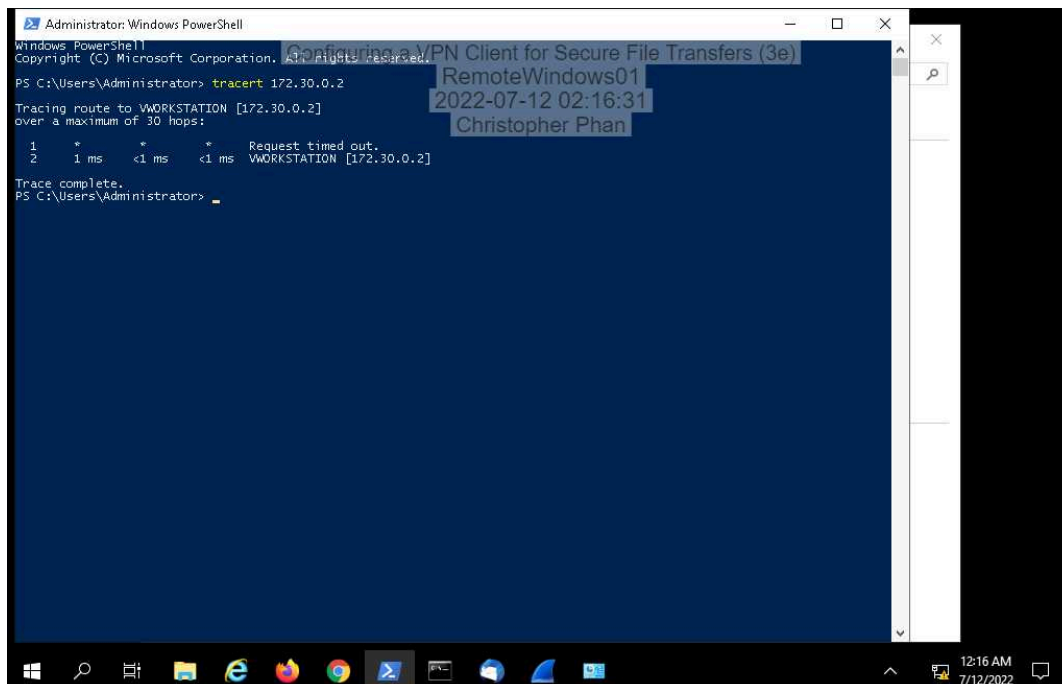
Section 2: Applied Learning

Part 1: Configure a Windows VPN Client

19. Make a screen capture showing the IPsec VPN connection encrypted with AES 256.



23. Make a screen capture showing your successful tracer to the remote machine.

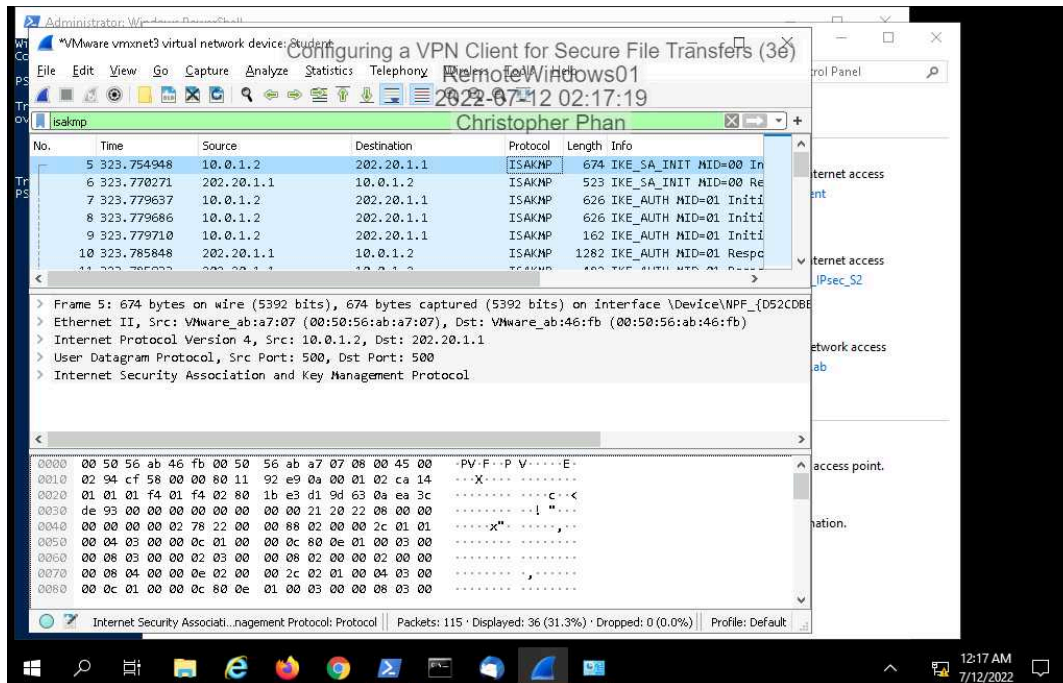


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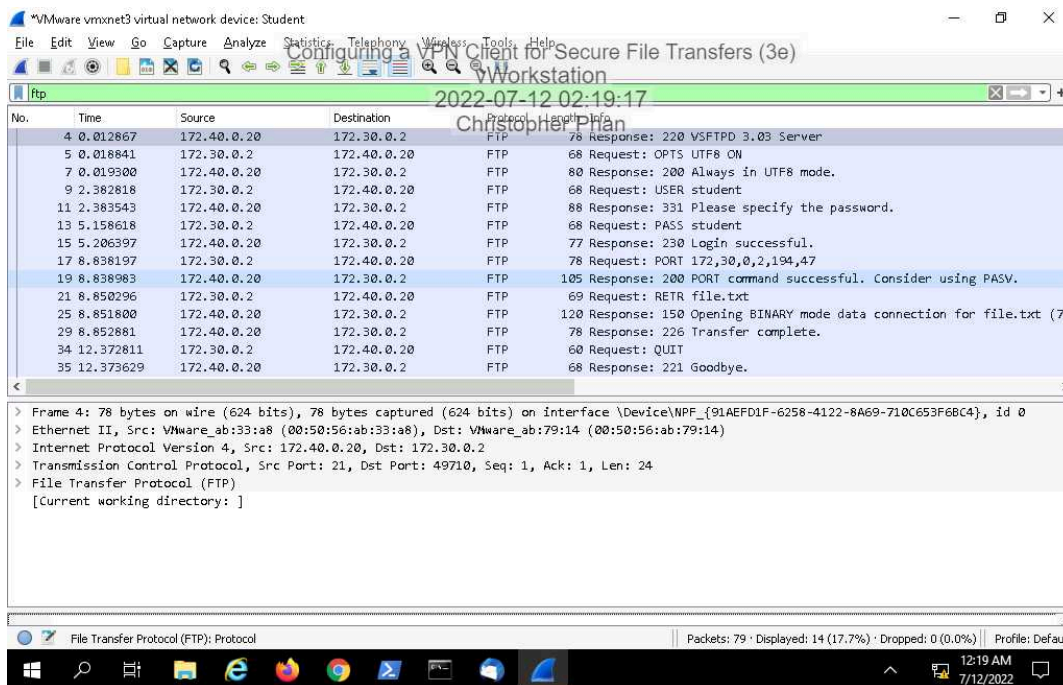
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Part 2: Compare Secure and Non-Secure File Transfers in Wireshark

7. Make a screen capture showing the IKE_SA_INIT and IKE_AUTH packets.



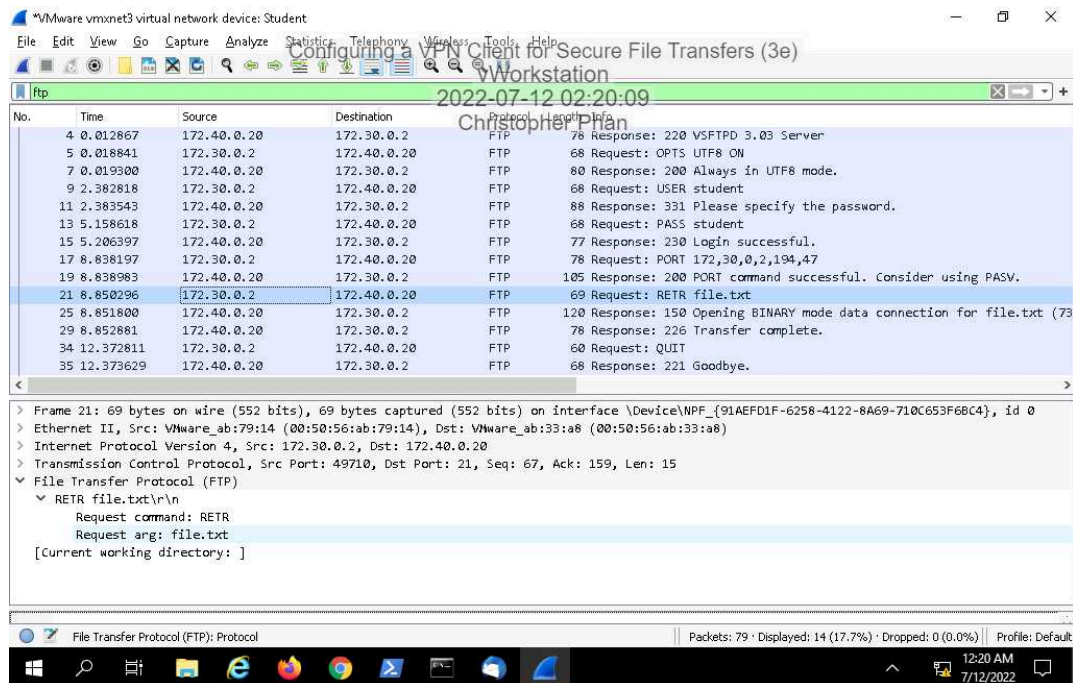
22. Make a screen capture showing the filtered FTP packets in your capture file.



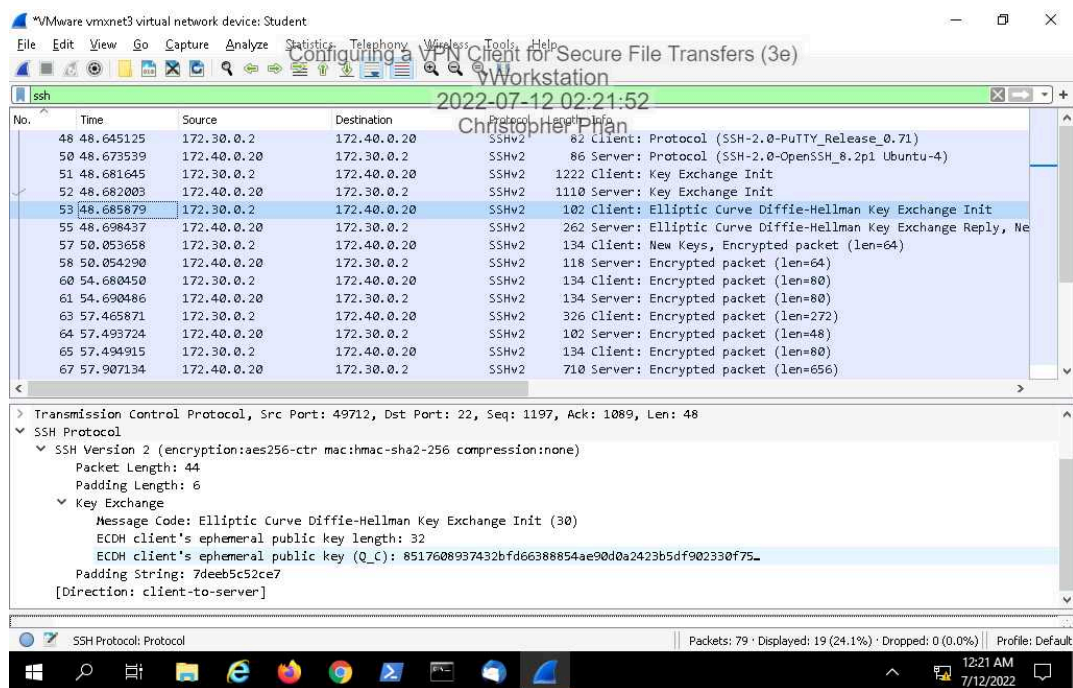
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24. Make a screen capture showing the contents of the file.txt file in the packet bytes pane.



27. Make a screen capture showing the filtered SSH packets in your capture file.



Section 3: Challenge and Analysis

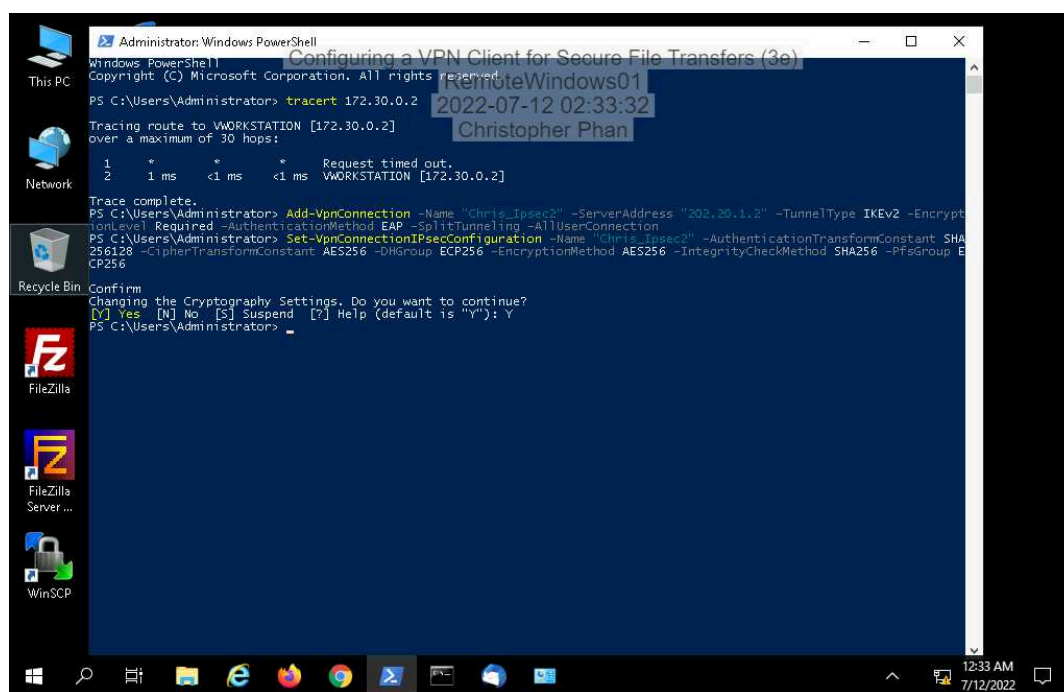
Part 1: Create a New VPN Connection using PowerShell

Document the command you used to add your VPN connection.

```
Add-VpnConnection -Name "Chris_IPsec2" -ServerAddress "202.20.1.2" -TunnelType IKEv2 -EncryptionLevel Required -AuthenticationMethod EAP -SplitTunneling -AllUserConnection
```

Part 2: Implement a Custom IPsec Policy

Make a screen capture showing the successfully executed `Set-VpnConnectionIPsecConfiguration` command in PowerShell.

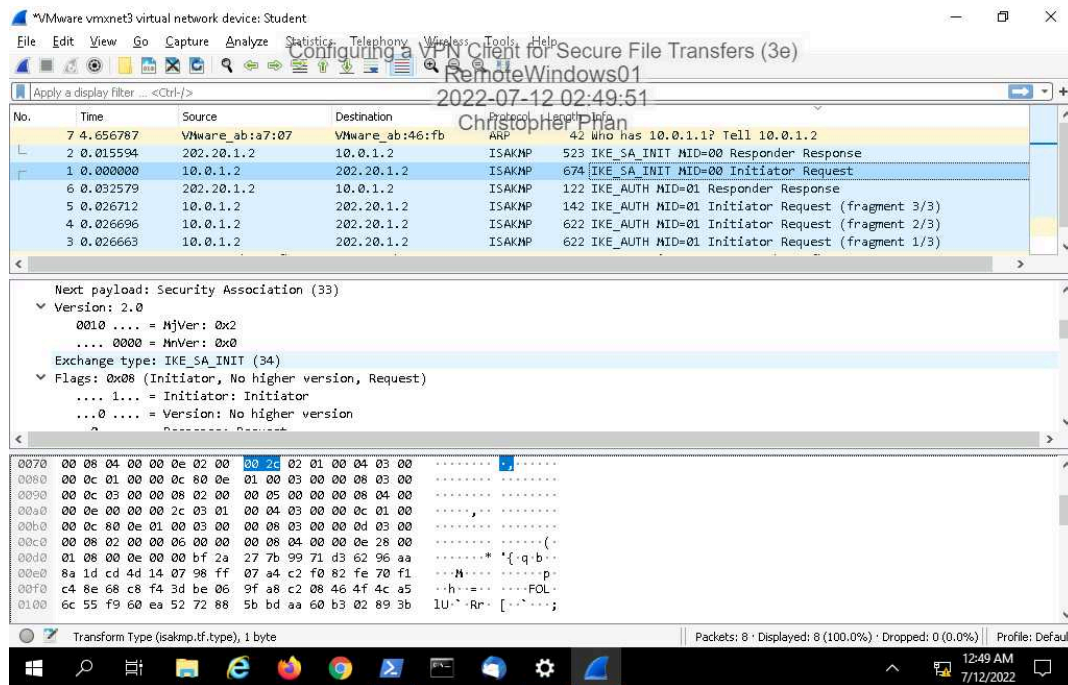


Part 3: Verify Your VPN Implementation using Wireshark

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Make a screen capture showing the **CREATE_CHILD_SA** exchange.



Make a screen capture showing the selected Diffie-Hellman transform.

